9 Jack bought a new boat for £12 500

The value, £V, of Jack's boat at the end of n years is given by the formula

$$V = 12\,500 \times (0.85)^n$$

(a) At the end of how many years was the value of Jack's boat first less than 50% of the value of the boat when it was new?

50% of £12500 = £6250

After 4 years: $12500 \times 0.85^4 = £6525.078125$

After 5 years: $12500 \times 0.85^5 = £5546.316406...$

1 mark
Evaluate the formula for whole numbers of years

The value decreases each year; it first falls below £6250 after 5 years.

1 mark
Correct first year

5

(2)

A savings account pays interest at a rate of R% per year.
Jack invests £5500 in the account for one year.

At the end of the year, Jack pays tax on the interest at a rate of 40%.
After paying tax, he gets £79.20

(b) Work out the value of R.

After 40% tax, Jack keeps 60% of the interest. P = percentage.

P	GBP
60	79.2
100	132

(÷0.6 on P; ÷0.6 on GBP)

1 mark
Recover the interest before tax: £132

GBP	P
5500	100
55	1
132	2.4

(÷100 on GBP; ÷100 on P; ×2.4 on GBP; ×2.4 on P)

1 mark
£132 is 2.4% of £5500

Interest before tax = £79.20 ÷ 0.6 = £132

R = 132 ÷ 5500 × 100 = 2.4

1 mark
Correct annual interest rate

2.4

(3)

(Total for Question 9 is 5 marks)